rsna-knee-abnormality-detection

5th Place Solution

Rank	5
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Topic ID	752190
Version	Original

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Retrieved with Kaggle CLI on 2026-10-27.

Thanks to RSNA and Kaggle for hosting this competition.

Thank you to my rock star team ✨ @tonylica , ✨ @tuckerarrants , ✨ @tomirol

1. Solution Overview

Because only 58 training studies had expert labels, we focused on constructing training labels for the remaining studies.

We trained study-level CNNs on soft labels derived from three sources:

- Language-model interpretations of radiology reports.
- Vision-model interpretations of selected MRI slices.
- Out-of-fold predictions from earlier models.

We used the 58 expert-labelled studies as a reference to calibrate (adjust) the soft labels used to train our models.

Our main models combined full-field MRI series with focused crops of the knee joint, medial and lateral compartments, and patellofemoral joint.

A bone-segmentation model placed the crops. The full-field series show the whole knee, and the crops show each region at up to twice the resolution.

The final ensemble first combined predictions from three model families—two EfficientNetV2-M families (M1, M2) and one ConvNeXt Base family (M3)—using a weighted probability average.

We then blended in separately trained region specialists (M4), four models fine-tuned for Lateral Meniscus (M5), and a ConvNeXt-Tiny family (M6). These later blends operated on prediction ranks, with components selected separately for each finding.

2. The problem: supervision from reports

The training set has 4,407 knee MRI studies, each with 3–14 series of the same knee in different planes and contrasts. The task is to predict 12 findings (ligament and meniscus injuries, osteoarthritis in three compartments, effusion, synovitis, Baker's cyst, bone contusion and fracture) on about 1,300 test studies, scored by mean AUC.

Only 58 training studies have expert labels. For the other 4,349, labels had to come from free-text radiology reports in several languages, and the test set has no reports. Reports make imperfect labels in three ways.

1. Missing information is not a negative label. One of the two public report-label tables we started from gave no answer for 27% of study–finding pairs, and for 84% of the synovitis pairs.

2. Report wording does not match the competition's definitions. The host's guide (topic 733343) counts an MCL injury only if it is an acute high-grade partial or complete tear, yet 199 low-grade MCL reports carried positive labels. On the expert-labelled studies, 0 of 5 such cases were positive (9 of 13 for other report-positive MCL cases), and only 10 of 22 small effusions were positive (24 of 28 for other report-positive effusions).

3. The label values were too high, and not by the same amount. "Yes" labels around 0.75, which came from hedged or low-grade report findings, were correct only 38% of the time. Synovitis inferred from an effusion got

the same label as findings the report stated directly (0.84). Experts confirmed it far less often: 38% against 65%. Because AUC depends on ranking, this uneven error puts studies in the wrong order.

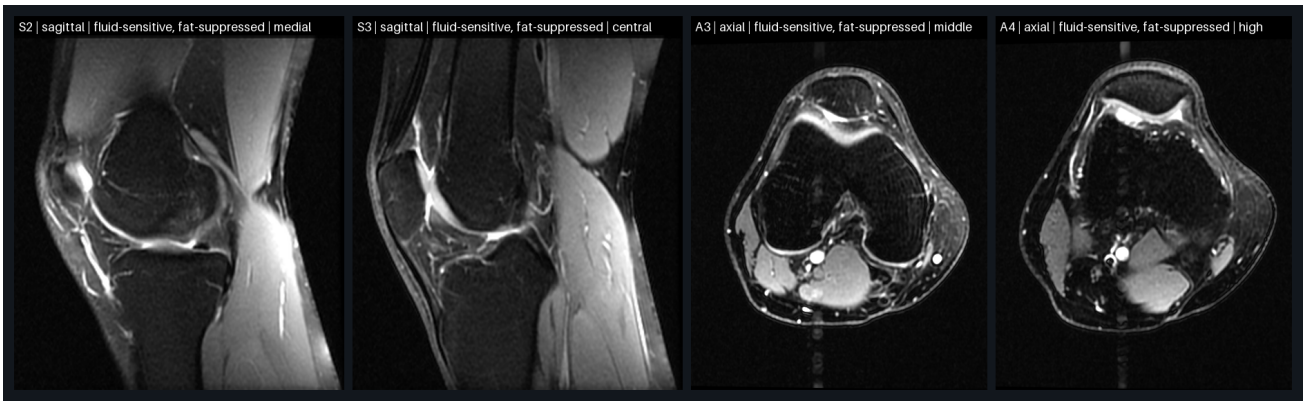
We used the 58 expert-labelled studies to correct the labels (section 3). Because they also guided our choices, scores on them are optimistic.

3. Training labels

We trained our CNNs using soft labels: values between 0 and 1 for each supervised finding. We developed these labels in three stages, retaining the original expert labels for the 58 expert-labelled studies.

1. Calibrate the public report labels. We started from two public tables containing language-model interpretations of the radiology reports, shared by Pilkwang and steven. Using the expert-labelled studies as a reference, we adjusted the soft labels to better match how often experts marked each finding positive. For each validation fold, we fitted these adjustments using only expert-labelled studies from the other folds. We call the resulting label set calA.

2. Obtain new interpretations of the reports and images. We processed every report with Claude Opus 5.5. For effusion, synovitis and Baker's cyst, gpt-6.1-sol also examined a panel of 10 fat-suppressed MRI slices per study, selected using bone masks (section 4). Combining the report and image interpretations produced label set F. Two further rounds of rereading disputed cases produced F2 label table.



Four of the slices the vision model reads for effusion, synovitis and Baker's cyst.

3. Combine these labels with predictions from earlier CNNs. We combined F2 with out-of-fold predictions from our CNNs to create v3. This added image-based predictions to the supervision derived from reports and selected slices. The later v5 labels combined F2, predictions from a CNN trained on v3, and predictions from an earlier ensemble containing region specialists. The blend parameters were fitted using the expert-labelled studies outside each validation fold. Our regional models M1 and M3 trained on v5.

Effect on private AUC (measured after the competition)

- **Report-label calibration:** CoAtNet improved from **0.916 to 0.926**, with training and inference unchanged.
- **Blended labels and longer training:** EfficientNetV2-M improved from **0.946 to 0.954** after switching from calA to v5 and training for 25 epochs instead of 17. Because both changed, we cannot attribute the gain to labels alone.

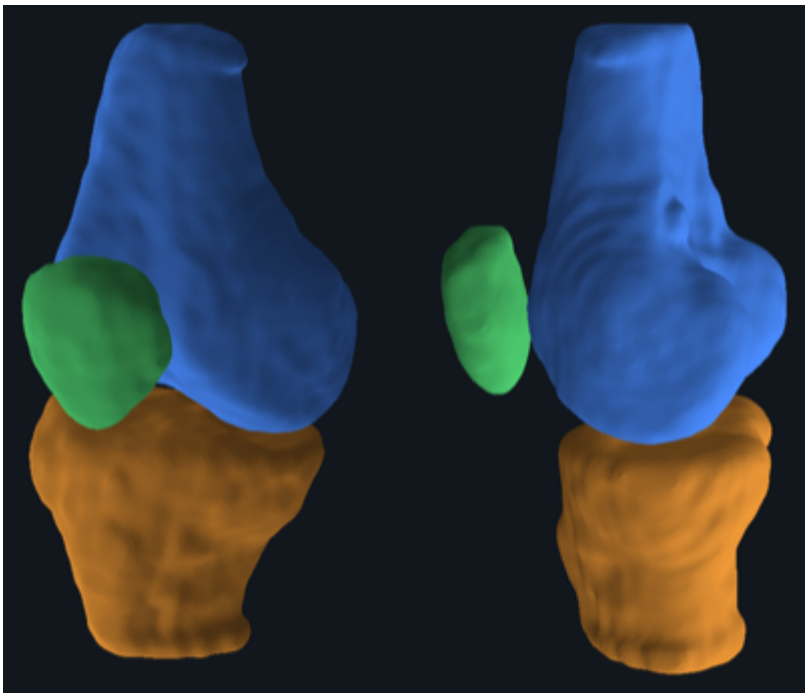
Validation caveat. A teacher CNN generating a training label could have trained on studies in the next model's validation fold. Its predictions could therefore pass information from validation into training. We treated local

scores as development diagnostics and used the public leaderboard to assess transfer to unseen studies (section 8).

4. Preprocessing and anatomy-guided crops

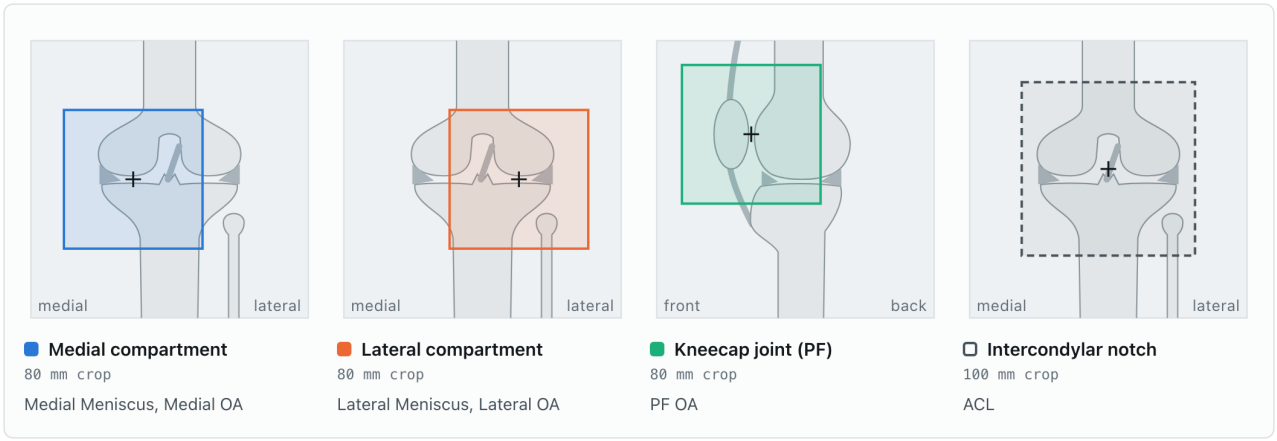
Full images. Each series is stored as 32 slices, spread evenly over the central 96% of the stack. Each slice is cut to the central 150 mm and resized to 384 × 384 px (about 0.39 mm per pixel). Intensities are scaled per series between the 1st and 99th percentile (99.9th for fat-suppressed series). Right knees are mirrored into a common left-knee orientation when the preprocessing code can determine the side. At test time the main models skip axial series without fat suppression, which gave a small local gain.

Finding the bones. We trained a bone-segmentation model (nnU-Net, initialised from TotalSegmentator-MRI) on 1,200 competition MRI series with agent-assisted femur, tibia and patella masks. A separate 3D CNN predicts whether the knee is left or right. Both models run at test time.



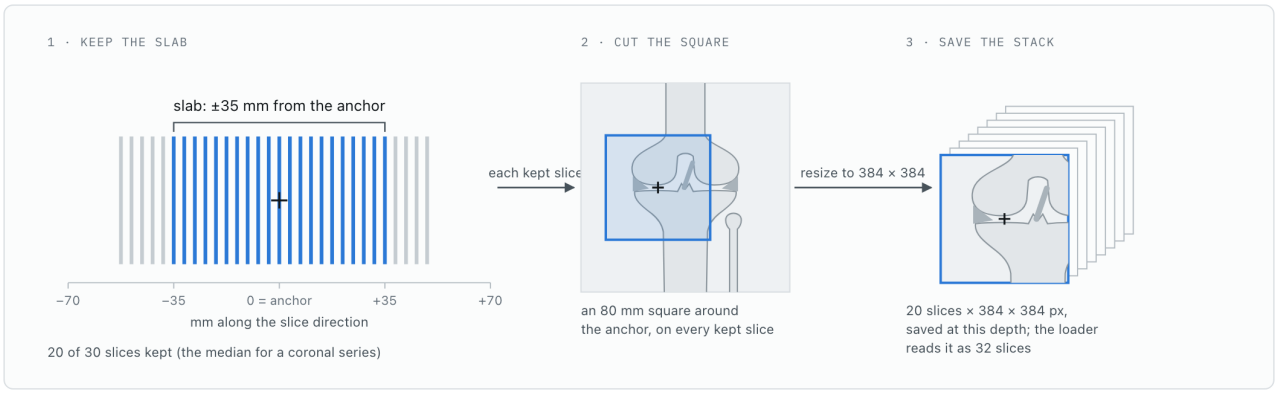
Bone segmentation of one knee, two views: femur (blue), tibia (orange), patella (green).

Four crop regions. Full-field images provide the overall anatomical context. The crops give smaller regions more pixels within the same input size, helping preserve local detail during resizing. From the bone masks we place a centre point (anchor) for each of four regions and cut a square around it.



The four crop locations and the findings that motivated their selection. Crosses mark the anchors; these associations do not restrict which views the main model can use for each prediction.

How one crop is made. A crop keeps the slices near its anchor, extracts a square from each slice, and resizes it to 384 × 384 pixels. An 80 mm crop represents about 0.21 mm per pixel, compared with 0.39 mm per pixel for the 150 mm full-field image. The same anatomical structure therefore occupies nearly twice as many pixels along each dimension.



One medial crop from a coronal series: the slices within ±35 mm of the anchor are kept.

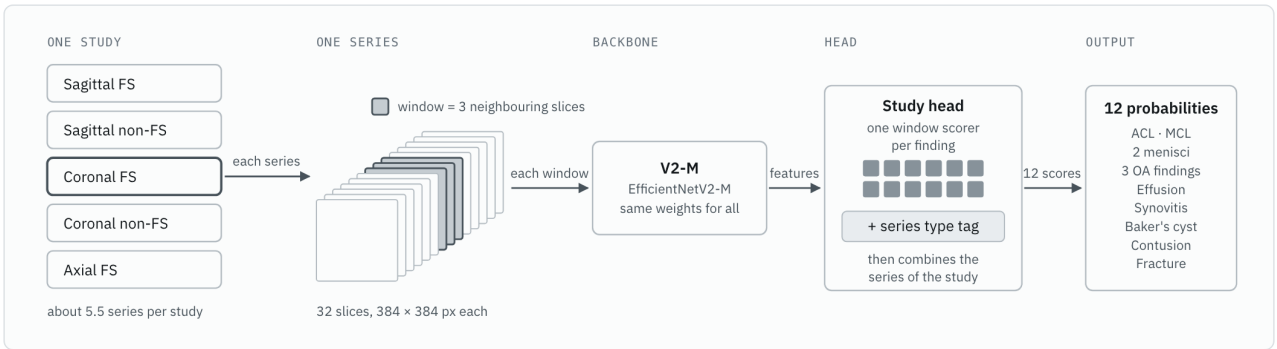


A real example: a full sagittal fat-suppressed image (left) and its 100 mm joint crop (right).

What the model receives. Adding the crop volumes increases the average number of input series from about 5.5 to 17.6. Each crop volume enters the model as an additional series, with a tag identifying its anatomical region and MRI sequence (section 5). The model predicts all 12 findings from the combined views. MCL, effusion, synovitis, Baker's cyst, contusion and fracture have no dedicated crop among these four regions, but can still use the full-field images and any relevant information within the crops. If a crop cannot be produced, the model uses the remaining available views. Separately trained region specialists provide additional predictions in the final ensemble (section 6).

5. Main model

Our main models share one design, a 2.5D CNN: a 2D backbone reads three-slice windows from the full images and from the four bone-anchored crops of section 4, and a small head combines them into 12 study-level probabilities.



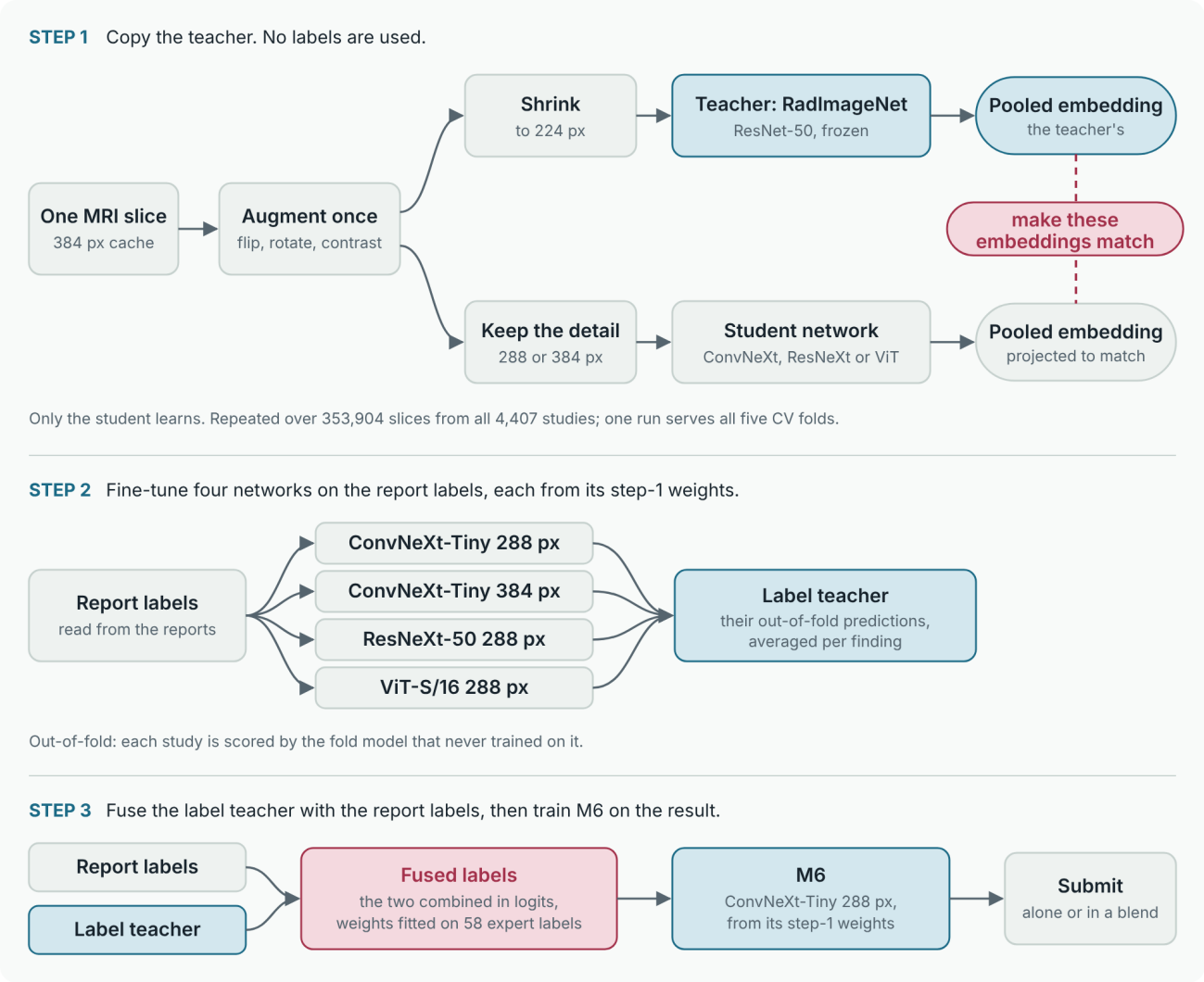
How one study is read, shown for the full images. The crops of section 4 enter the same way, as extra series.

- **Windows.** Each series, full image or crop, is cut into windows of three neighbouring slices. The backbone reads each window as one RGB image, with the same weights for every series.
- **Head.** For each finding, the head weights the windows inside a series, then the series inside the study. A learned tag tells it what each series is: 60 types (12 plane-and-contrast types for each of the full image and the four crop regions).
- **Pretraining.** Before the competition data, each backbone learned to predict 327 yes/no readings derived from radiologists' MOAKS/WORMS grades, on 2,414 knees from the Osteoarthritis Initiative (OAI). In two single-model tests, this start added 0.001 to the public score.
- **Training.** Site folds (section 8), 25 epochs, AdamW, an exponential moving average of the weights, and random crops, rotations and intensity jitter, but no flips, because left and right matter.
- **Inference.** 15 windows per series, the centre 336 px of each 384 px slice, and the mean of the five fold models.

The final ensemble uses four models built this way or close to it:

Model	Backbone	Reads	Folds	Labels (section 3)	Private alone
M1	EfficientNetV2-M	full images + 4 crops	5	v5	0.957
M2	EfficientNetV2-M	full images + 4 crops	4	half LLM-only, half M1 predictions	not submitted
M3	ConvNeXt Base	full images + 4 crops	5	v5	0.958
M6	ConvNeXt-Tiny	one slice at a time	5	teacher-fused labels	0.947

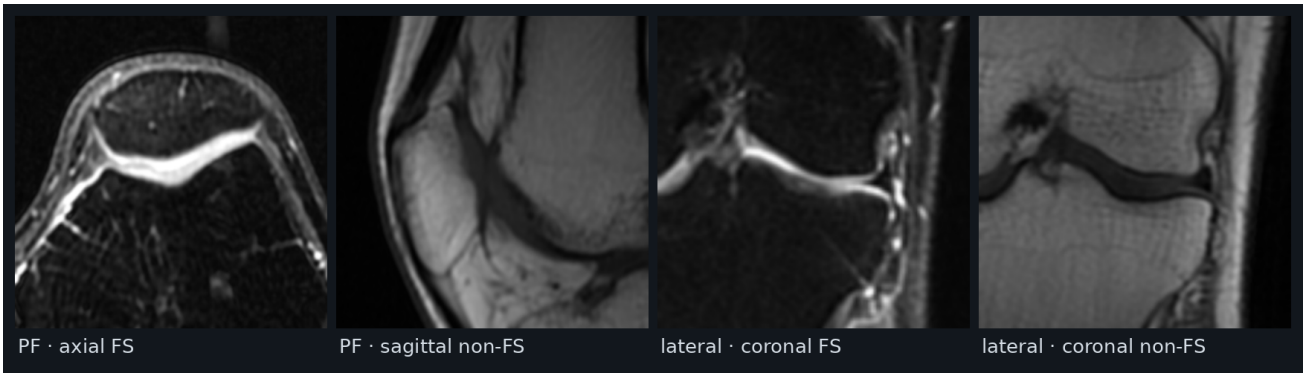
M6, our ConvNeXt-Tiny family, uses a different design. It reads one slice at a time at 288 px (16 slices per series, sorted into 9 plane-and-contrast slots), scores every slice, and only then pools. Its backbone first learned to copy a frozen RadImageNet ResNet-50 without any labels, and was then fine-tuned on the 12 findings, then again on teacher fused labels like the main model.



M6's three training steps: copy a frozen RadImageNet teacher, then learn the 12 findings, then retrain on teacher-fused labels.

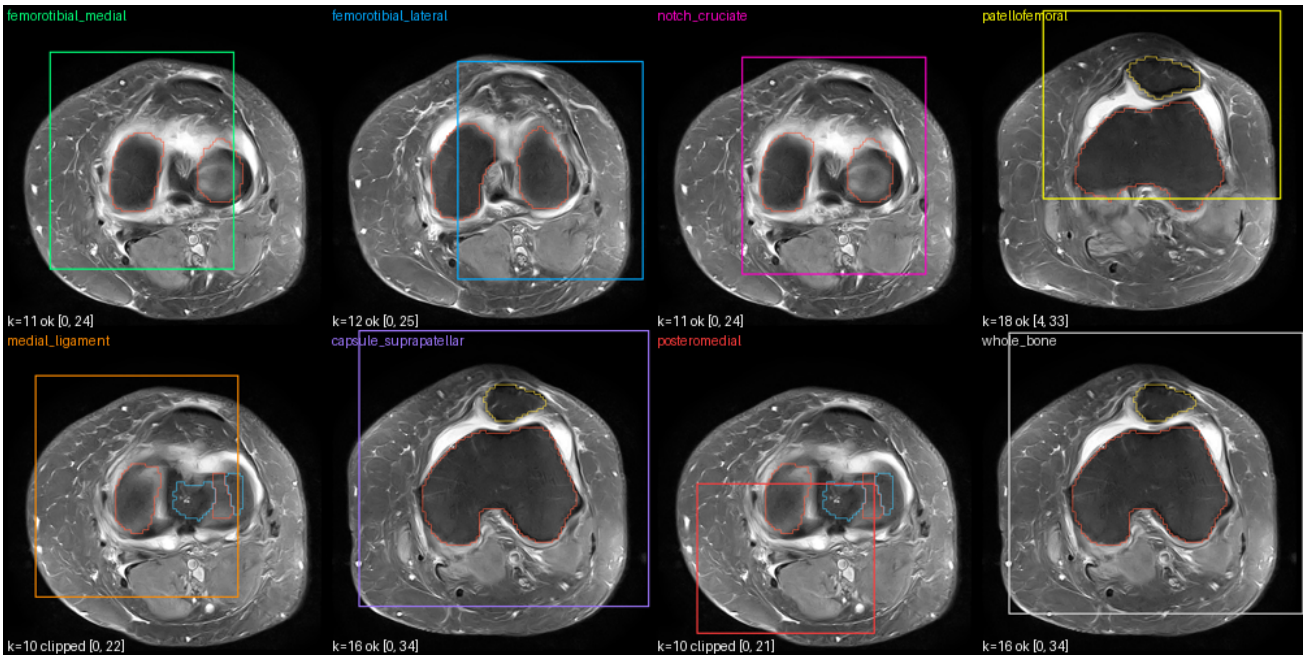
6. Region specialists and targeted additions

What a specialist is. A region specialist (M4) is one of our main-model networks, fine-tuned for 6 more epochs on the crops of one knee region only. We built seven kinds. Each fold model starts from the same fold of one of our models (ConvNeXt V2 Tiny, ConvNeXt Small or EfficientNetV2-M). A specialist predicts all 12 findings, but the blend reads only the findings assigned to it.



Specialist crops, 80 mm at 384 px: the patellofemoral joint (axial fat-suppressed, sagittal non-fat-suppressed) and the lateral compartment (coronal fat-suppressed and non-fat-suppressed).

The specialists add two things to the main model's crops. They also read the non-fat-suppressed series, which was the largest single gain for PF OA. And they cover regions the main model does not crop: a 120 mm bone crop (contusion, fracture), the posteromedial corner (Baker's cyst), and a taller bone crop for effusion and synovitis.



Region boxes placed from the bone masks, one axial slice each (our anatomy code).

Each finding reads one or two specialists:

Finding	Specialist crop
ACL, Lateral Meniscus	lateral compartment
MCL, Medial Meniscus	medial compartment
Medial OA	medial compartment + notch
Lateral OA	notch
PF OA	patellofemoral
Contusion	bone + notch
Fracture	bone
Effusion, Synovitis	taller bone crop
Baker's cyst	bone + posteromedial

Two routes look odd (ACL from the lateral crop, Lateral OA from the notch crop). Both won pre-registered checks on held-out folds.

How they are blended. For each finding: 0.5 × rank of the main models + 0.5 × rank of its specialists. The weight 0.5 won a leave-one-fold-out search.

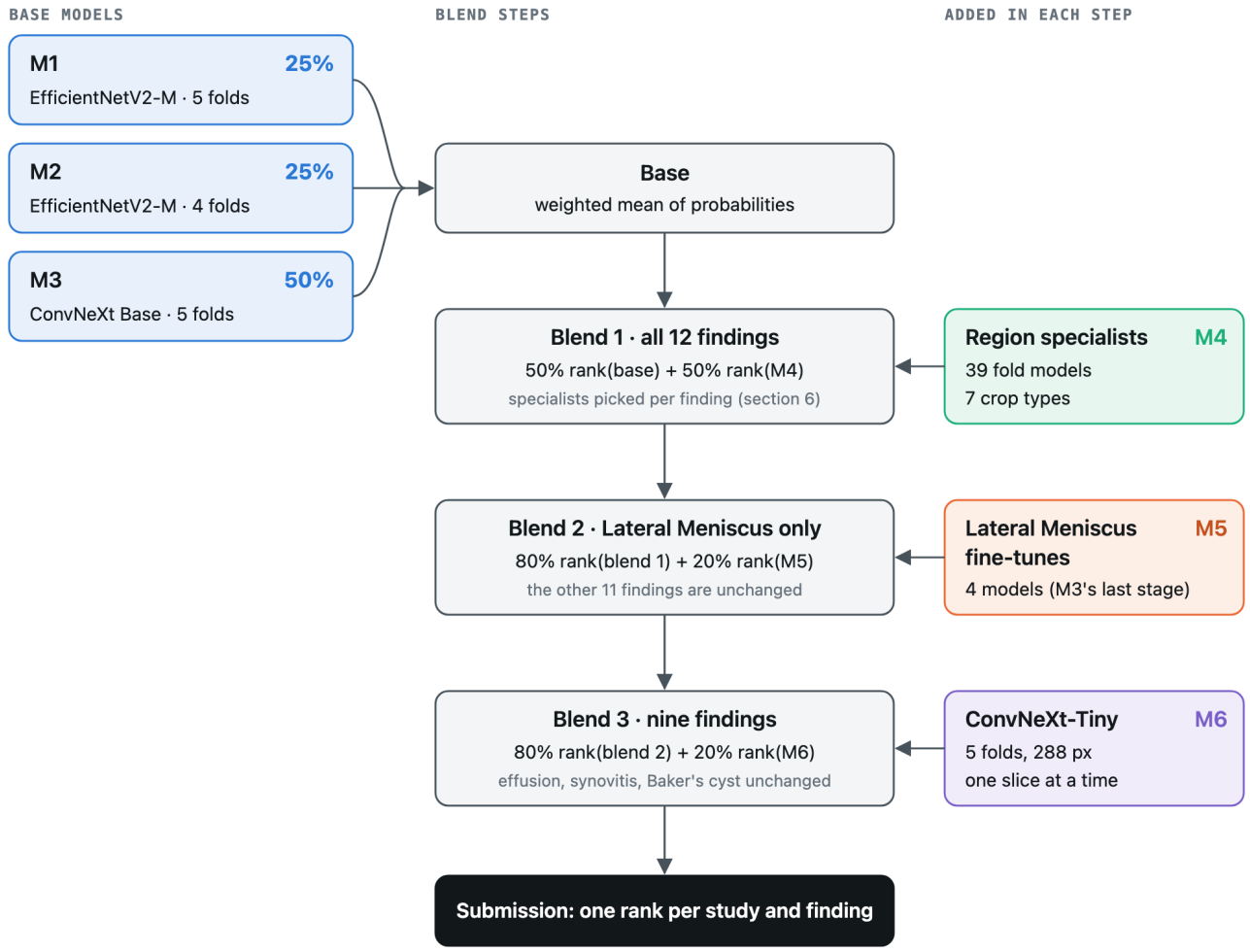
What they added. On a weaker base the specialists added 0.003 public AUC (0.963 → 0.966) and 0.002 private. On our regional models they added about 0.001 or less, because M1's own crops already carry most of their signal.

Two targeted additions in the final ensemble.

- **Taller bone crop for effusion and synovitis.** The old bone crop held only 78–80% of the slices cited for these two findings (the missed slices were all above it), so its axial slab was extended 20 mm upward.
- **Lateral Meniscus fine-tunes (M5).** Of 40 short fine-tunes of M3 aimed at weak findings, only the Lateral Meniscus one passed. It retrains M3's last stage and head on explicit report statements and is blended at 20% rank. It reuses M3's computation for the earlier layers, so it costs little time.

7. The final ensemble

Our final submission first averages three models, then applies three rank blends. Each blend works one finding at a time: it ranks the test studies and mixes the ranks.



The final ensemble. Each blend step ranks the studies again for each finding, so the step weights cannot be combined into one average of the original probabilities.

- **Base.** 25% M1 + 25% M2 + 50% M3, averaged as probabilities. One kernel decodes each study and places the crops once for all three models.

- **Blend 1, all 12 findings.** Half the base rank, half the rank of the specialists (M4) assigned to that finding (section 6).
- **Blend 2, Lateral Meniscus only.** 80% of blend 1's rank, 20% of the rank of the four Lateral Meniscus fine-tunes (M5).
- **Blend 3, nine findings.** 80% of blend 2's rank, 20% of M6's rank. Effusion, synovitis and Baker's cyst skip this step.

AUC only reads the order of the studies within each finding, so ranks give models with differently calibrated probabilities a common scale. The weights were fixed by simple rules or local fold checks, not tuned on the leaderboard.

On the private leaderboard this ensemble scored 0.958. The same ensemble without its last three additions (the taller bone crop, M5 and M6) scored 0.958 and 0.957 in two runs.

8. Validation and results

How we validated.

- **Site folds.** Scanner settings reveal the hospital, and a model can learn a hospital's base rates instead of the knee. We grouped the studies into 20 hospital-like clusters and kept each cluster inside one of the five folds. When a study's site was also in training, the same model scored 0.019 higher local AUC against the report labels, but not higher on the 58 expert-labelled studies.
- **Local scores.** We tracked two, and neither could pick between label sets:
  - Cross-validation AUC against the report labels. It rewards agreeing with the reports, not with the experts (section 2). It also runs high for models trained on teacher-blended labels, because those labels came partly from teachers that had trained on the validation fold (section 3). B1 and B2 (the first two rows of the table below) scored the same here (0.873), but B2 scored higher on both leaderboards (public +0.007, private +0.008).
  - AUC on the 58 expert-labelled studies. It uses the right definitions, but 58 studies is a small sample, and we had already used them for label and blend decisions, so it runs high too.

Where the score gains came from:

Model	Change from the row above	Public	Private
EfficientNetV2-M, calibrated report labels (B1)	—	0.960	0.946
Same model, teacher-blended labels (v5), 25 epochs instead of 17 (B2)	labels and training length	0.967	0.954
Same model and labels, plus crops of the joint, compartments and patellofemoral joint (M1)	anatomy-guided crops	0.969	0.957
ConvNeXt Base with the same crops and labels (M3)	backbone	0.969	0.958
Final ensemble (M1–M6)	ensemble, section 7	0.969	0.958

Better labels gave the largest step (+0.008 private) and the anatomy crops the second (+0.003). The other backbone, the specialists and the final blends added 0.001 or less, so the final ensemble ties our best single models.

9. Inference within the time limit

Kaggle allowed 9 hours on two T4 GPUs, without internet. Our single regional models ran in 2 h 19 min (M1) to 2 h 57 min (M3); the full ensemble took 8 h 18 min. Four things made it fit and kept it safe:

- **Shared preprocessing.** One kernel decodes each study and runs the bone segmentation and cropping once, then scores M1, M2, M3 and the Lateral Meniscus fine-tunes (M5) on the same inputs. This saves about 42% of a base model's GPU time. M5 also reuses M3's early layers.
- **Both GPUs.** One worker process per T4, each taking every other study.
- **A time budget per stage.** M6 gets 90 minutes, the base models must finish by 6.75 h, and the specialists stop at 8.25 h.

10. Code, weights and acknowledgements

Code and weights: coming soon.

Acknowledgements and references

- RSNA, the annotators and Kaggle for the competition and the data.
- The public report-label tables (CC0) our first labels were built on: Pilkwang's `report_labels_v2` and Steven's `llm_labels_v4_blend`.
- The public Claude Opus 4.5 label table by Tejas N (NTejas-1/RSNA-Knee-Abnormality-Detection, MIT), used in M6's report labels.
- The Osteoarthritis Initiative (OAI), whose knee MRI and MOAKS/WORMS readings pretrained our backbones:

> Data and/or research tools used in the preparation of this manuscript were obtained and analyzed from the controlled access datasets distributed from the Osteoarthritis Initiative (OAI), a data repository housed within the NIMH Data Archive (NDA). OAI is a collaborative informatics system created by the National Institute of Mental Health and the National Institute of Arthritis, Musculoskeletal and Skin Diseases (NIAMS) to provide a worldwide resource to quicken the pace of biomarker identification, scientific investigation and OA drug development. Dataset identifier(s): NDA Collection 2343, DOI 10.15154/64dx-3822.

- RadImageNet, the teacher network for M6's distillation.
- TotalSegmentator (MRI model) for the bone model's starting weights, nnU-Net for its architecture and training, and nnInteractive for drawing the bone masks.
- timm for the EfficientNetV2 and ConvNeXt backbones.
- Furlanello et al., Born Again Neural Networks (ICML 2018), the idea behind the teacher-fused labels of M6 and M2.

Supplemental Q&A

> **Optimo · 2026-10-24T16:17:37.313000**
>
> Congratulations to the whole team and to Rashmi for becoming a competition grandmaster!
>
> Could you guys just describe how you used Opus 5.5 and gpt-6.1-sol to get labels: is that through API calls where you prepare a specific prompt and give the selected slices? How did you pick the slices to show the LLM? How much did it cost (approximately)? Or is it through a plan, if so how do you prompt your model to go through all the cases ? Also how did you wrote the prompts? How many version have you tried? How can you decide which version is better (GT58 only?) ?

> **Rashmi B · 2026-10-24T21:51:25.540000**
>
> Thank you @optimo !
>
> API calls would be quite expensive. My labeling attempts didn't do well, @tomirol 's did and should have more to add to this.
>
> You can use `claude -p` or `codex exec` to run in headless mode … even better you can ask an agent to launch these for you it manages the whole thing - hundreds of subagents across claude and codex. I ran out of my weekly quota 😅 , thankfully I had resets. Have agents write prompts also - much better and detailed for subagents to label.

> **matheus · 2026-10-25T12:12:30.670000**
>
> Hi @optimo, thanks for the questions. We basically just used Claude Code and Codex directly (no API), with a slice selection using the bone geometry we segmented beforehand. It seems most teams did something similar in one way or another, and we did not use a lot of quota: only a few hours for a full labeling and 20-30% of codex weekly usage at high effort.